



Republic of the  
Philippines  
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# IT123 NETWORK CONFIGURATION AND MANAGEMENT

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# IT 123 - WEEK 5 LABORATORY

## Lab Objectives

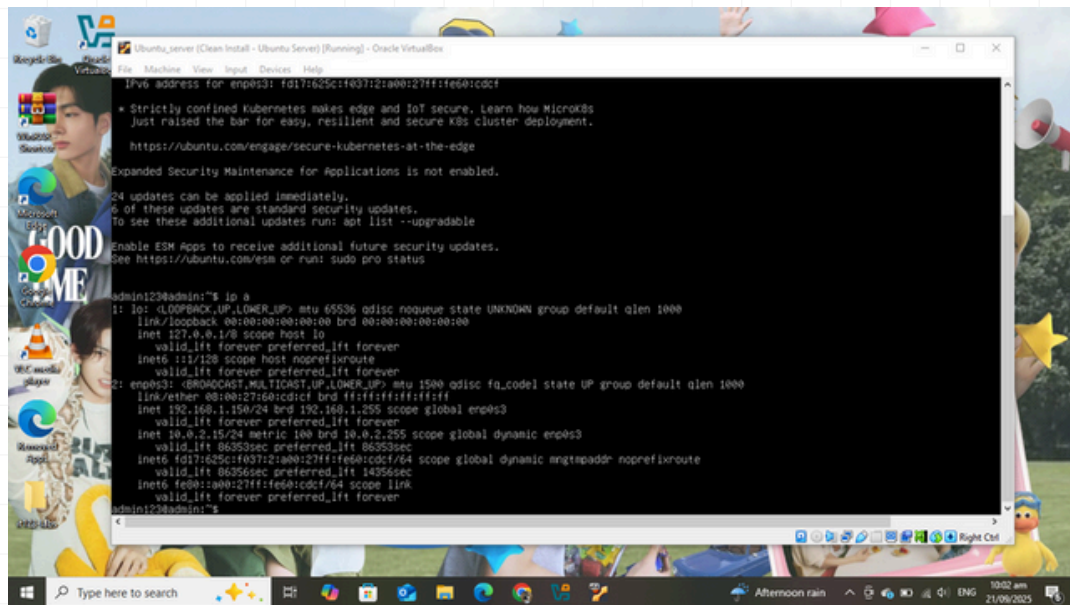
By the end of this lab, students will be able to:

1. Configure static and dynamic IP addresses in Linux (Ubuntu Server).
2. Verify network connectivity and troubleshoot using CLI tools.
3. Understand basic network configuration in Windows (optional).
4. Apply security practices by testing connectivity and limiting access.

## Part 1 – Checking Current Network Configuration (Ubuntu Server)

1. Show active interfaces:

ip a

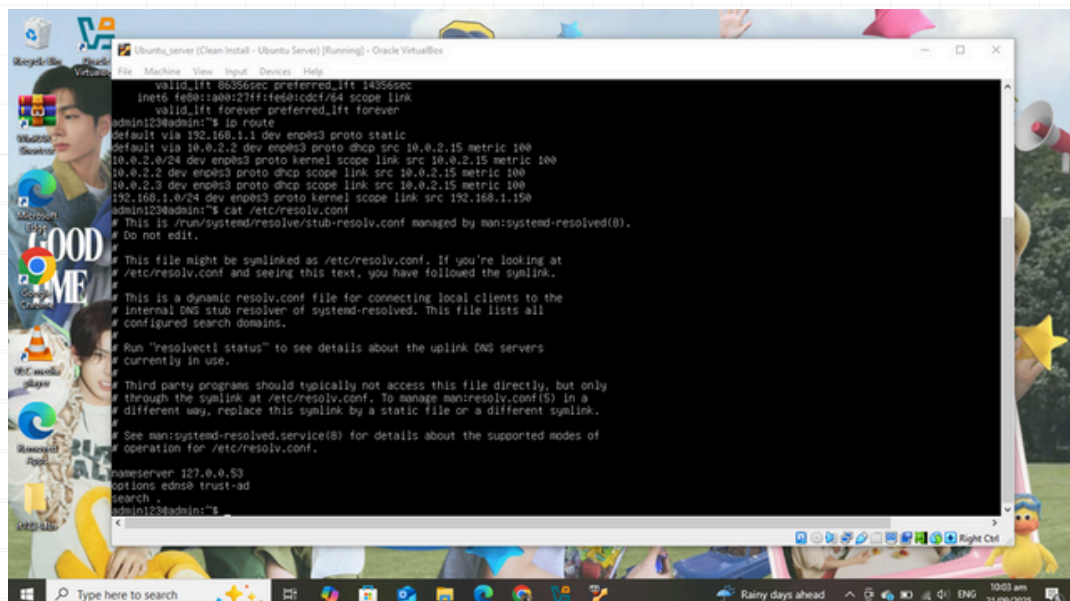


```
admin123@admin1:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BRIDGE,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:6a:cd:c1f brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.150/24 brd 192.168.1.255 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86353sec preferred_lft 86353sec
    inet6 fd17:625c:f637:2:a00:27ff:fe60:cdcf/64 scope global dynamic mngtaddr: noprefixroute
        valid_lft 86353sec preferred_lft 14356sec
    inet6 fe80:a00:27ff:fe60:cdcf:64 scope link
        valid_lft forever preferred_lft forever
admin123@admin1:~$
```

2. Check default route and DNS:

ip route

cat /etc/resolv.conf



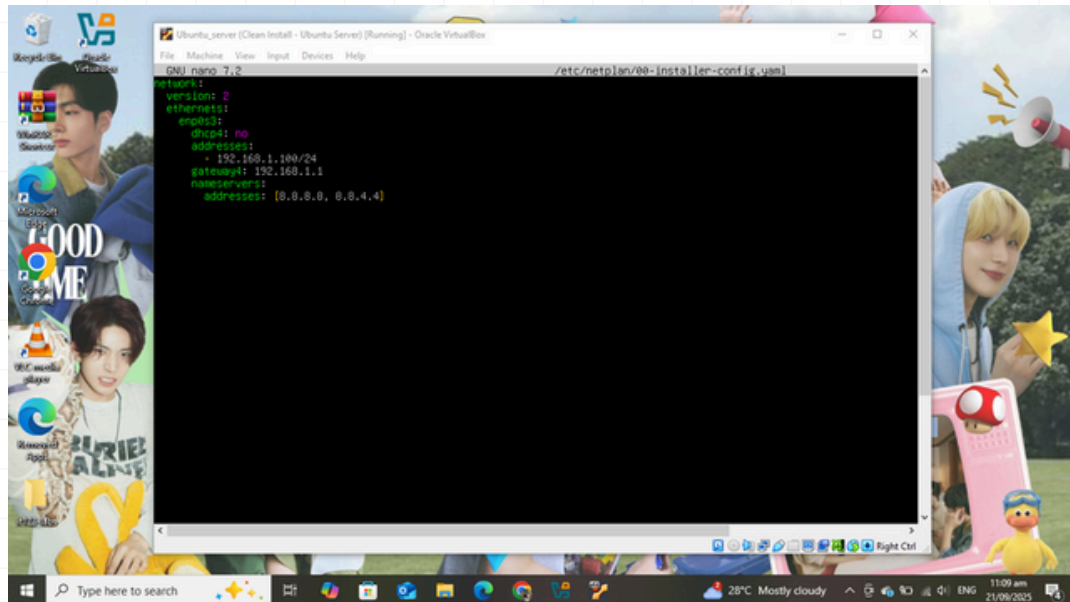
```
admin123@admin1:~$ ip route
default via 192.168.1.1 dev enp0s3 proto static
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
10.0.2.2 dev enp0s3 proto dhcp scope link src 10.0.2.15 metric 100
10.0.2.3 dev enp0s3 proto dhcp scope link src 10.0.2.15 metric 100
192.168.1.0/24 dev enp0s3 proto kernel scope link src 192.168.1.150
admin123@admin1:~$ cat /etc/resolv.conf
# This is /run/systemd/resolve/stub-resolv.conf managed by man:systemd-resolved(8).
# Do not edit.
#
# This file might be symlinked as /etc/resolv.conf. If you're looking at
# /etc/resolv.conf and seeing this text, you have followed the symlink.
#
# This is a dynamic resolv.conf file for connecting local clients to the
# internal DNS stub resolver of systemd-resolved. This file lists all
# configured search domains.
#
# Run "resolvectl status" to see details about the uplink DNS servers
# currently in use.
#
# Third party programs should typically not access this file directly, but only
# through the symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a
# different way, replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.
nameserver 127.0.0.53
options edns0 trust-ad
search .
admin123@admin1:~$
```

## Part 2 – Configuring a Static IP Address (Ubuntu Server)

Ubuntu 20.04+ uses netplan for network configs.

1. Open the netplan config file:

```
sudo nano /etc/netplan/00-installer-config.yaml
```



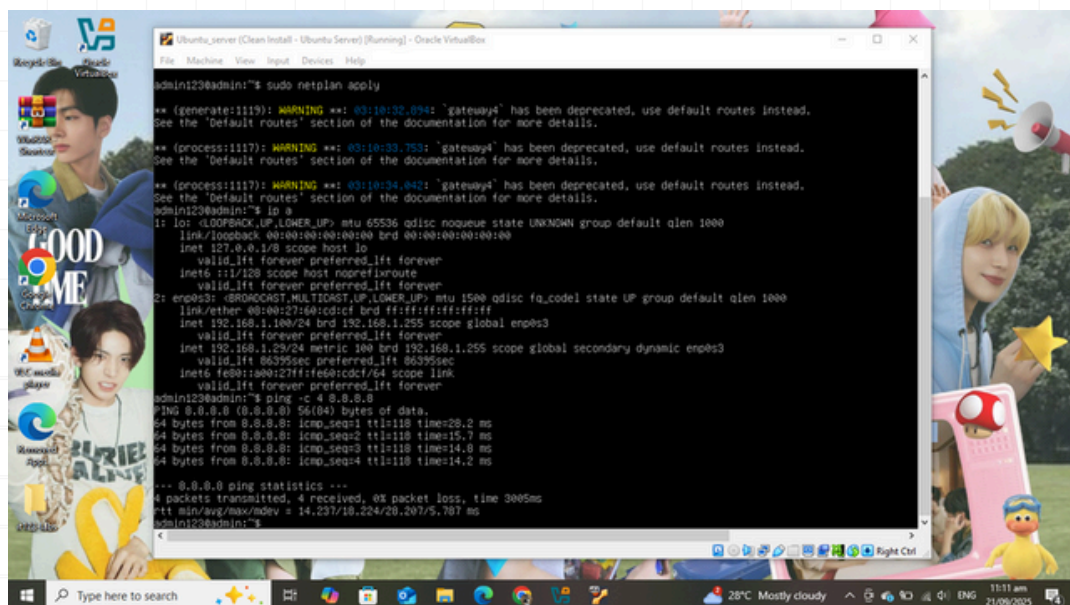
2. Apply changes:

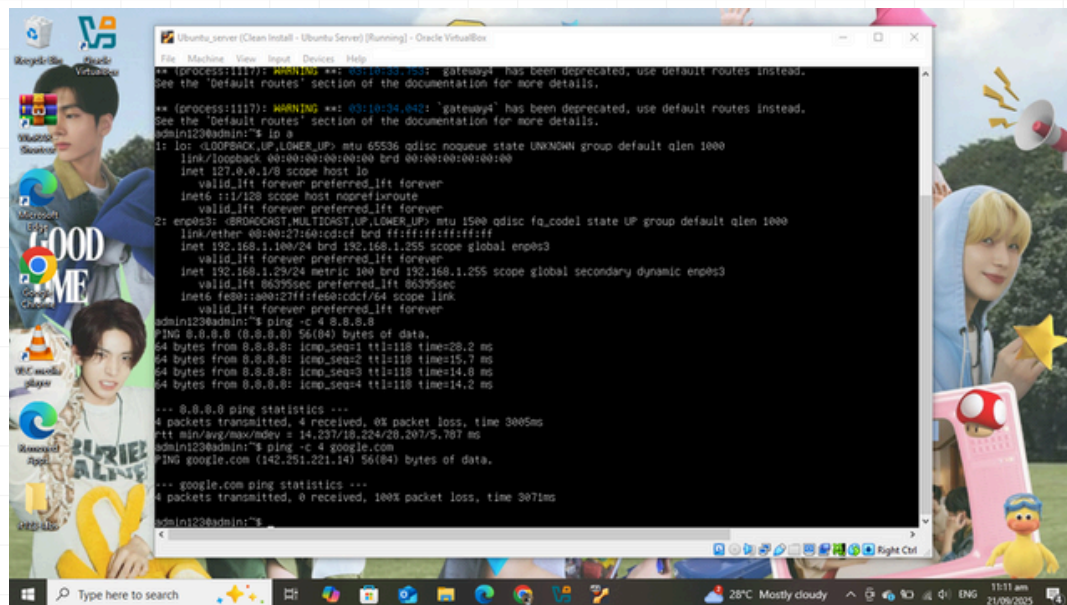
`sudo netplan apply` and Verify:

`ip a`

`ping -c 4 8.8.8.8`

`ping -c 4 google.com`





```
File Machine View Input Devices Help
** (process:1177): WARNING **: 00:10:23.123: 'gateway4' has been deprecated, use default routes instead.
See the 'Default routes' section of the documentation for more details.

** (process:1177): WARNING **: 00:10:34.042: 'gateway4' has been deprecated, use default routes instead.
See the 'Default routes' section of the documentation for more details.

admin123@admin1:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:16:01:cd brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.100/24 brd 192.168.1.255 scope global ens3
        valid_lft forever preferred_lft forever
    inet 192.168.1.29/24 metric 100 brd 192.168.1.255 scope global secondary dynamic ens3
        valid_lft 86306sec preferred_lft 86306sec
    inet6 fe80::a00:27ff:fe60:cdcf/64 scope link
        valid_lft forever preferred_lft forever
admin123@admin1:~$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=118 time=28.2 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=118 time=15.7 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=118 time=14.8 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=118 time=14.2 ms

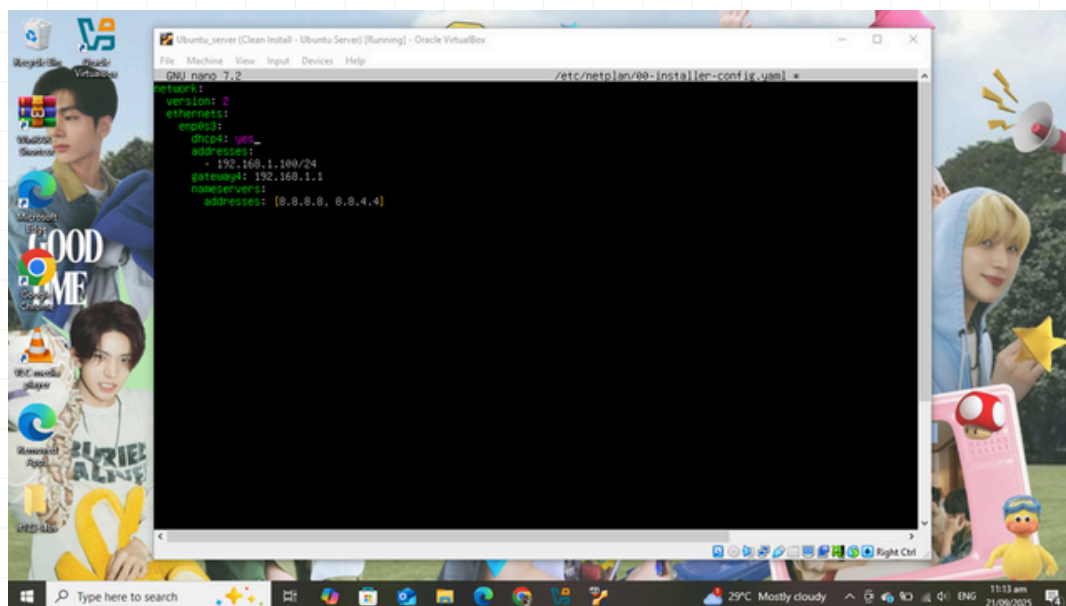
--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 300ms
rtt min/avg/max/mdev = 14.237/18.224/28.207/5.787 ms
admin123@admin1:~$ ping -c 4 google.com
PING google.com (142.251.221.14) 56(84) bytes of data:
4 packets transmitted, 0 received, 100% packet loss, time 507ms

admin123@admin1:~$
```

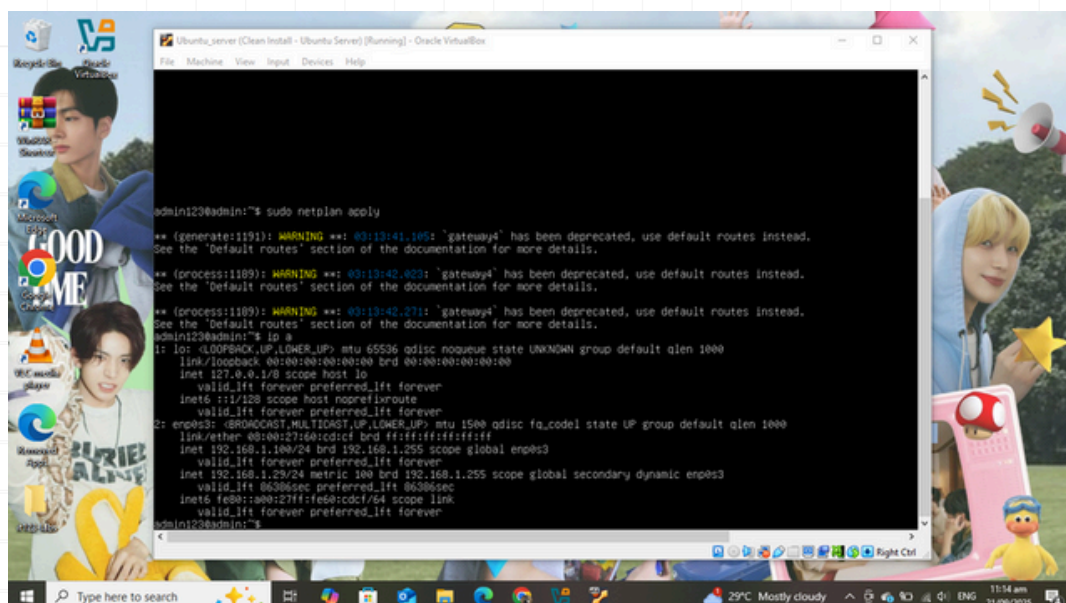
## Part 3 – Switching Back to DHCP (Ubuntu Server)

1. Edit the netplan config file again:

dhcp4: yes; Apply changes: sudo netplan apply; Verify: ip a



```
File Machine View Input Devices Help
GNU nano 2.9.2 /etc/netplan/00-installer-config.yaml
network:
  version: 2
  ethernet:
    ens3:
      dhcp4: yes
      addresses:
        - 192.168.1.100/24
      gateway: 192.168.1.1
      nameservers:
        addresses: [8.8.8.8, 8.8.4.4]
```



```
File Machine View Input Devices Help
admin123@admin1:~$ sudo netplan apply
** (generate:1191): WARNING **: 00:13:41.145: 'gateway4' has been deprecated, use default routes instead.
See the 'Default routes' section of the documentation for more details.

** (process:1189): WARNING **: 00:13:42.023: 'gateway4' has been deprecated, use default routes instead.
See the 'Default routes' section of the documentation for more details.

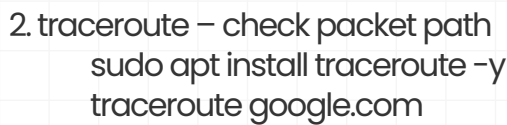
** (process:1189): WARNING **: 00:13:42.271: 'gateway4' has been deprecated, use default routes instead.
See the 'Default routes' section of the documentation for more details.

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:16:01:cd brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.100/24 brd 192.168.1.255 scope global ens3
        valid_lft forever preferred_lft forever
    inet 192.168.1.29/24 metric 100 brd 192.168.1.255 scope global secondary dynamic ens3
        valid_lft 86306sec preferred_lft 86306sec
    inet6 fe80::a00:27ff:fe60:cdcf/64 scope link
        valid_lft forever preferred_lft forever
admin123@admin1:~$
```

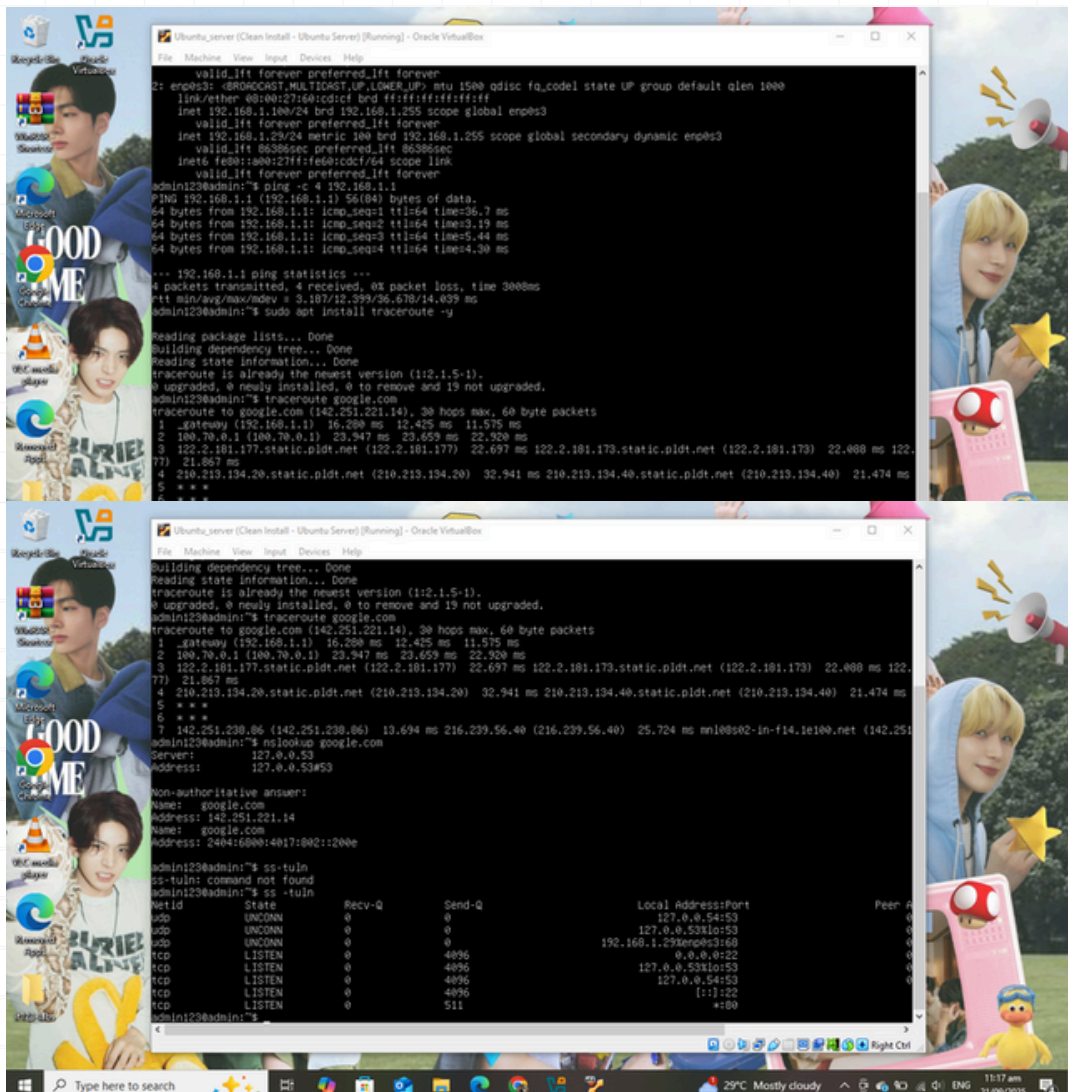


1.ping – test connectivity  
ping -c 4 192.168.1.1

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ping -c 4 192.168.1.1



```
2. traceroute – check packet path
sudo apt install traceroute -y
traceroute google.com
```



3. nslookup – test DNS resolution  
nslookup google.com
4. netstat/ss – check active connections  
ss -tln

```

Building dependency tree... Done
Reading state information... Done
traceroute is already the newest version (1:2.1.5-1).
0 upgraded, 0 newly installed, 0 to remove and 19 not upgraded.
admin123@admin123:~$ traceroute google.com
traceroute to google.com (142.251.221.14), 30 hops max, 60 byte packets
 1 gateway (192.168.1.1)  16.200 ms  12.425 ms  11.515 ms
 2 190.70.0.1 (190.70.0.1)  23.947 ms  23.659 ms  22.928 ms
 3 122.2.101.177.static.pldt.net (122.2.101.177)  22.697 ms  122.2.101.173.static.pldt.net (122.2.101.173)  22.408 ms  122.2.101.173.static.pldt.net (122.2.101.173)  21.867 ms
 4 210.213.134.20.static.pldt.net (210.213.134.20)  32.941 ms  210.213.134.40.static.pldt.net (210.213.134.40)  21.474 ms
 5 * * *
 6 * * *
 7 142.251.238.86 (142.251.238.86)  13.694 ms  216.239.56.40 (216.239.56.40)  25.724 ms  mni08s02-in-f14.1e100.net (142.251.238.86)
admin123@admin123:~$ nslookup google.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.221.14
Name:   google.com
Address: 2404:6800:4017:802::200e

admin123@admin123:~$ ss -tln
ss-tln: command not found
admin123@admin123:~$ ss -tln
State      Recv-Q    Send-Q    Local Address:Port    Peer Address
tcp        UNCONN    0         0                    127.0.0.1:54153
tcp        UNCONN    0         0                    127.0.0.1:53
tcp        UNCONN    0         0                    192.168.1.29:8080
tcp        LISTEN    0         4096               0.0.0.0:22
tcp        LISTEN    0         4096               127.0.0.1:53
tcp        LISTEN    0         4096               127.0.0.1:54153
tcp        LISTEN    0         4096               [::]:22
tcp        LISTEN    0         511              *:*

```

## Part 6 – Student Exercise (Ubuntu CLI Focus)

### Scenario:

Your company requires that one Ubuntu server has a static IP (192.168.1.150) for internal services, while another should remain on DHCP for flexibility.

### Tasks:

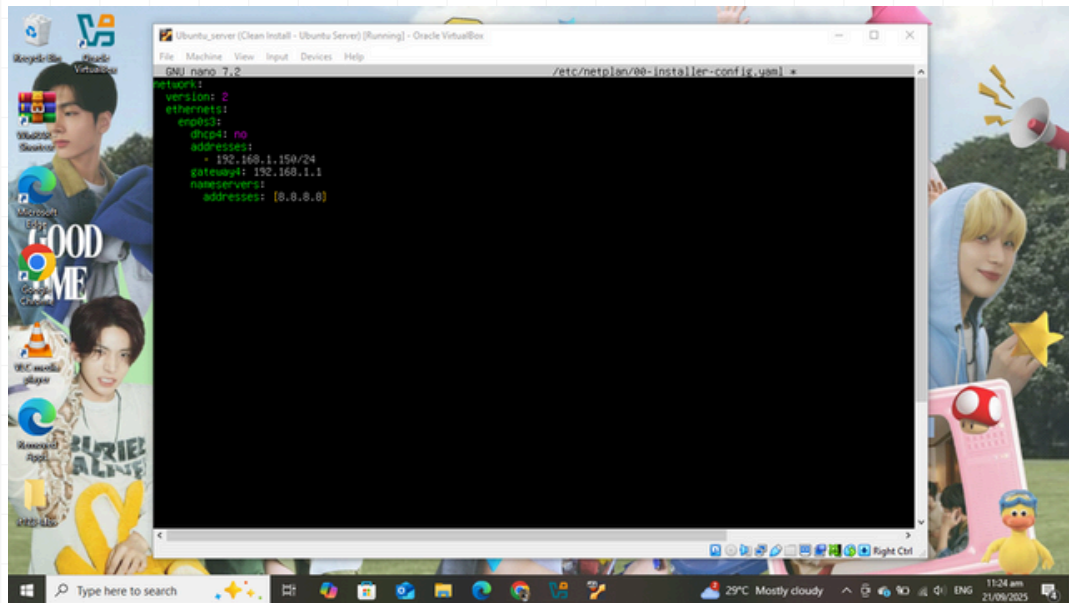
1. Configure Server A with static IP: 192.168.1.150/24
  - Gateway: 192.168.1.1
  - DNS: 8.8.8.8
2. Configure Server B with DHCP enabled.
3. Verify both servers can:
  - Ping each other.
  - Ping the internet (e.g., google.com).
4. Document steps and include screenshots/command outputs.

### OUTPUT:

#### 1. Configure Server A with Static IP

To configure Server A with a static IP address of 192.168.1.150, follow these steps:

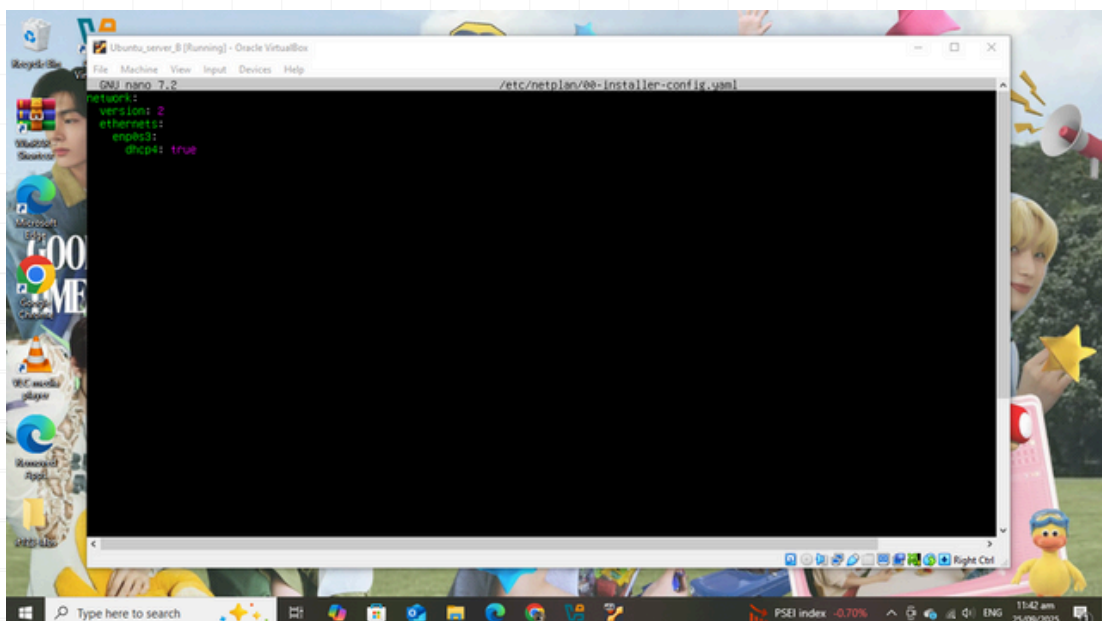
- Open the terminal on Server A.
- Edit the Netplan configuration file using the following command  
sudo nano /etc/netplan/00-installer.yaml



## 2. Configure Server B with DHCP

For Server B, the configuration to use DHCP is straightforward:

- Open the terminal on Server B.
- Edit the Netplan configuration file as follows  
`sudo nano /etc/netplan/00-installer.yaml`  
Set the configuration to enable DHCP

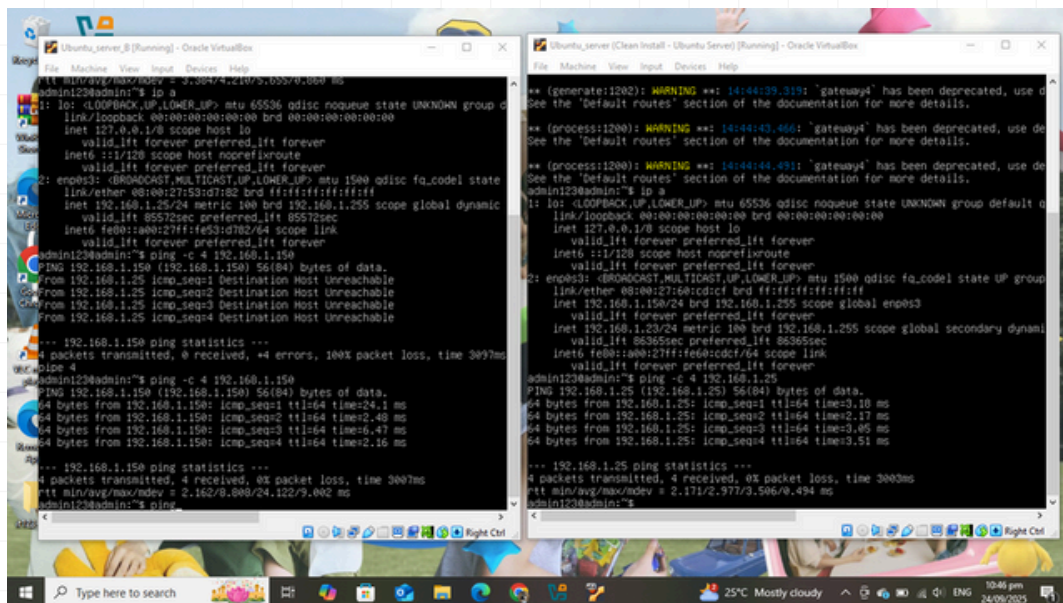




### 3. Verify Connectivity

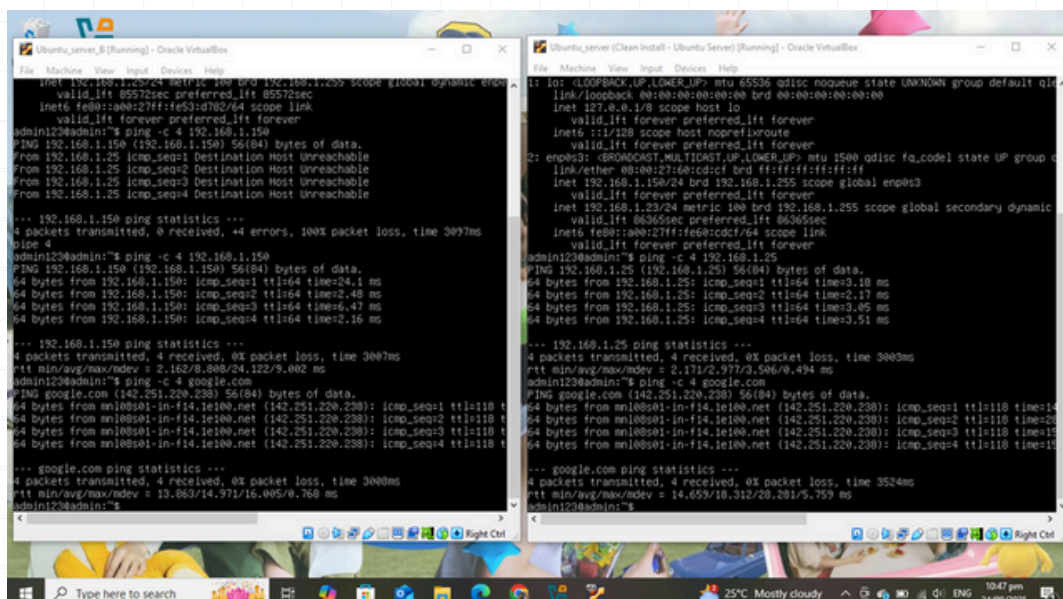
To ensure both servers can communicate with each other and access the internet, perform the following tests:

- Ping Server A from Server B:  
ping -c 4 192.168.1.150
- Ping Server B from Server A:  
ping -c 4 192.168.1.25



The image shows two terminal windows from Oracle VM VirtualBox. The left window, titled 'Ubuntu\_server\_B [Running]', shows the output of a ping command from 192.168.1.150 to 192.168.1.25. The output indicates that the connection is successful, with 4 packets transmitted, 4 received, and 0% packet loss. The right window, titled 'Ubuntu\_server [Clean Install - Ubuntu Server] [Running]', shows the output of a ping command from 192.168.1.25 to 192.168.1.150. The output also indicates a successful connection with 4 packets transmitted, 4 received, and 0% packet loss.

- Ping the Internet from both servers:  
Test the connection to an external address, such as Google:  
ping -c 4 google.com



The image shows two terminal windows from Oracle VM VirtualBox. The left window, titled 'Ubuntu\_server\_B [Running]', shows the output of a ping command from 192.168.1.150 to google.com. The output indicates a successful connection to google.com with 4 packets transmitted, 4 received, and 0% packet loss. The right window, titled 'Ubuntu\_server [Clean Install - Ubuntu Server] [Running]', shows the output of a ping command from 192.168.1.25 to google.com. The output also indicates a successful connection to google.com with 4 packets transmitted, 4 received, and 0% packet loss.